INCH-POUND
MS8004D
w/AMENDMENT 1
09 September 2005
SUPERSEDING
MS8004D
24 September 2003

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE, REUSABLE FITTINGS, HIGH TEMPERATURE, MEDIUM PRESSURE, FLANGE-TO-FLANGE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-25579.

Hose cut-off factor (HCOF) for assembly with class 1 fittings: See table I.

TABLE I. Hose cut-off factor (HCOF) for assembly with class 1 fittings. 1/

Style 2/ Corrosion resistant steel (CRES)	Fitting ends		HCOF (size vs length) 3/					
	1	2	8	10	12	16	20	24
A	MS27054	MS27054	2.46	2.84	2.76	3.22	3.38	3.62
B	MS27054	MS27056	2.48	2.84	3.28	3.59	3.91	4.31
C	MS27054	MS27058	2.44	2.83	3.30	3.66	4.03	4.49
D	MS27056	MS27056	2.50	2.84	3.80	3.96	4.44	5.00
E	MS27056	MS27058	2.46	2.83	3.82	4.03	4.56	5.18
F	MS27058	MS27058	2.42	2.82	3.84	4.10	4.58	5.36

1/ Class 1 fittings, styles A through F, are made from CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 23.50 inch length assembly of size 10, style F is calculated as follows: 23.50 - 2.82 = 20.68.

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HCOF for assembly with class 2 fittings: See table II.

TABLE II. HCOF for assembly with class 2 fittings. 1/

Styles 2/ (Aluminum and CRES)	Fitting ends		HCOF (size vs length) 3/					
	1	2	8	10	12	16	20	24
G	MS27054	MS27054	2.46	2.84	2.76	3.22	3.38	3.62
H	MS27054	MS27056	2.48	2.84	3.28	3.59	3.91	4.31
J	MS27054	MS27058	2.44	2.83	3.30	3.66	4.03	4.49
K	MS27056	MS27056	2.50	2.84	3.80	3.95	4.44	5.00
M	MS27056	MS27058	2.46	2.83	3.82	4.03	4.56	5.18
N	MS27058	MS27058	2.42	2.82	3.84	4.10	4.68	5.36

1/ Class 2 fittings, styles G through N, are made from combination of aluminum and CRES.
2/ See figure 1 for depiction of styles.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 23.50 inch length assembly of size 10, style F is calculated as follows: 23.50 - 2.82 = 20.68.

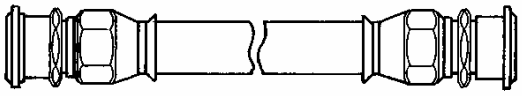
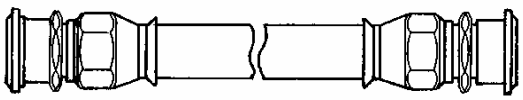
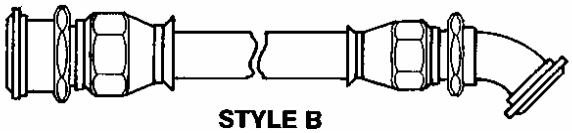
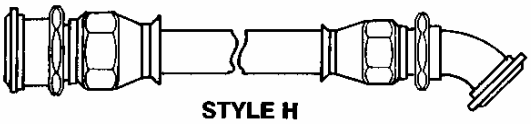
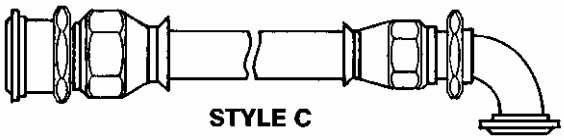
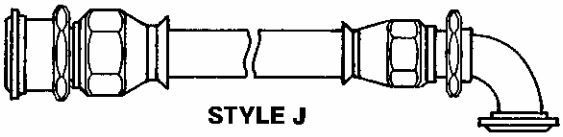
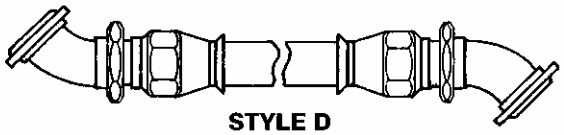
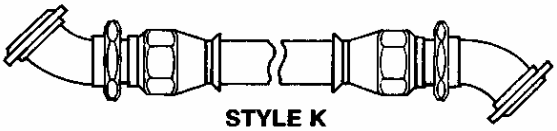
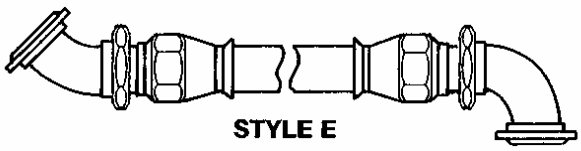
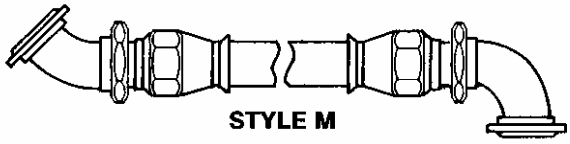
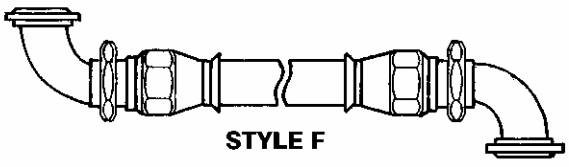
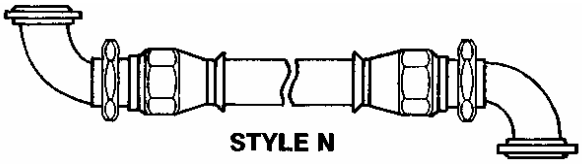
Protective sleeve code: See table III.

TABLE III. Protective sleeve code.

Code	Type
A	SAE AS1072 sleeve, fire protection, silicone covered, temperature ranging from -65°F to 450°F and intermittently to 500°F, secured with CRES bands as required. 1/
B	SAE AS1073 - code B sleeve, abrasion protection, heat shrinkable, black polyolefin, temperature ranging from -65°F to 250°F.
C	SAE AS1291 - code A sleeve, chafe guard, extruded seamless white PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
D	SAE AS1291 - code C sleeve, chafe guard, extruded seamless transparent FEP, temperature ranging from -65°F to 350°F, secured with CRES bands as required.
E	SAE AS1298 sleeve, heavy wall chafe guard, extruded seamless black PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
L	Lock-wire hole
F	Code A + L
G	Code B + L
H	Code C + L
J	Code D + L
K	Code E + L

1/ To prevent wicking of fluids, the cut end of the fire protective sleeve (code A) shall be coated with Room Temperature Vulcanized (RTV) silicone rubber prior to installation. After installation, cracks and voids in the fire protective sleeve shall be coated with RTV rubber to prevent exposure of fiberglass.

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Class 1 fittings (450°F) Corrosion resistant steel		Class 2 fittings (275°F) Combination aluminum and corrosion resistant steel	
End 1	End 2	End 1	End 2
ASSEMBLY LENGTH TYPICAL  STYLE A		ASSEMBLY LENGTH TYPICAL  STYLE G	
 STYLE B		 STYLE H	
 STYLE C		 STYLE J	
 STYLE D		 STYLE K	
 STYLE E		 STYLE M	
 STYLE F		 STYLE N	

NOTES:
1. See tables I and II for fitting ends and HCOF.
2. Fittings shall mate with the mounting pad as shown in MS33786.

FIGURE 1. Classes 1 and 2 hose assembly styles with flange-to-flange reusable fittings.

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REQUIREMENTS

Dimensions: Unless otherwise specified, all dimensions are in inches.

Fittings: All fittings shall be qualified in accordance with MIL-DTL-27272.

Hose: The hose shall be qualified in accordance with MIL-DTL-27267.

Assembly classification: Class 1 and class 2 hose assemblies, as specified in MIL-DTL-25579, have been incorporated into the Part or Identifying Number (PIN) as a part of styles (see tables I and II and figure 1).

Angular alignment: Hose assemblies with elbow fittings on each end shall have the angular orientation between the elbows measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o'clock, to the centerline of the other fitting (see figure 2). The elbow fitting drop height shall be as shown on figure 3 and table IV. When applicable, the angular alignment shall be expressed in three digits and specified in the PIN.

Protective sleeve: If required, the hose assembly shall include a protective sleeve (see table III) and its code shall be included in the PIN. Fire protective sleeve shall be subjected to testing in accordance with MIL-DTL-25579.

Assembly length: Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see MIL-DTL-25579); however, tolerances on the length of each hose assembly shall be as follows:

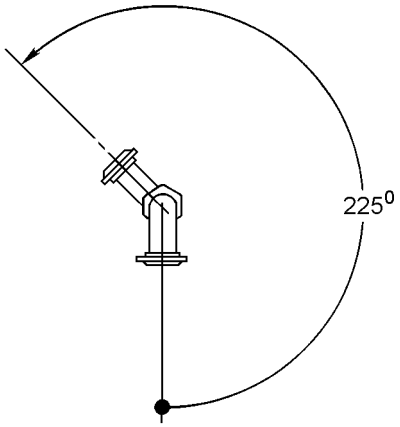
a. ± 1/8 inch for lengths under 18 inches.
b. ± 1/4 inch for lengths from 18 inches to 36 inches.
c. ± 1/2 inch for lengths from 36 inches to 50 inches.
d. ± 1% for lengths over 50 inches.

TABLE IV. Size code and elbow fitting drop height.

Size	Reference tube OD	Size code	Maximum drop height of elbow fitting	
			(a) 1/	(b) 1/
8	0.500	H	0.458	0.927
10	0.625	J	0.410	0.931
12	0.750	K	0.503	1.191
16Z	1.000	M	0.540	1.317
20Z	1.250	N	0.605	1.535
24Z	1.500	P	0.659	1.723

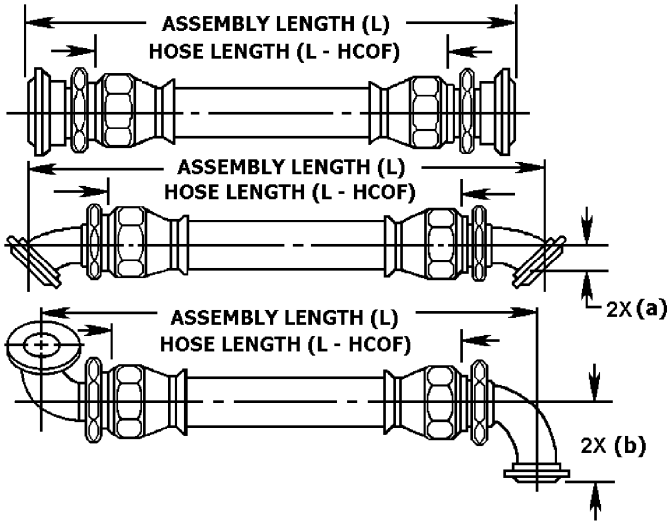
1/ Dimensions (a) and (b) are depicted on figure 3.

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NOTE: Angular alignment shall be measured in degrees with a tolerance of ±2°.

FIGURE 2. Measurement of angular alignment between elbow fittings.



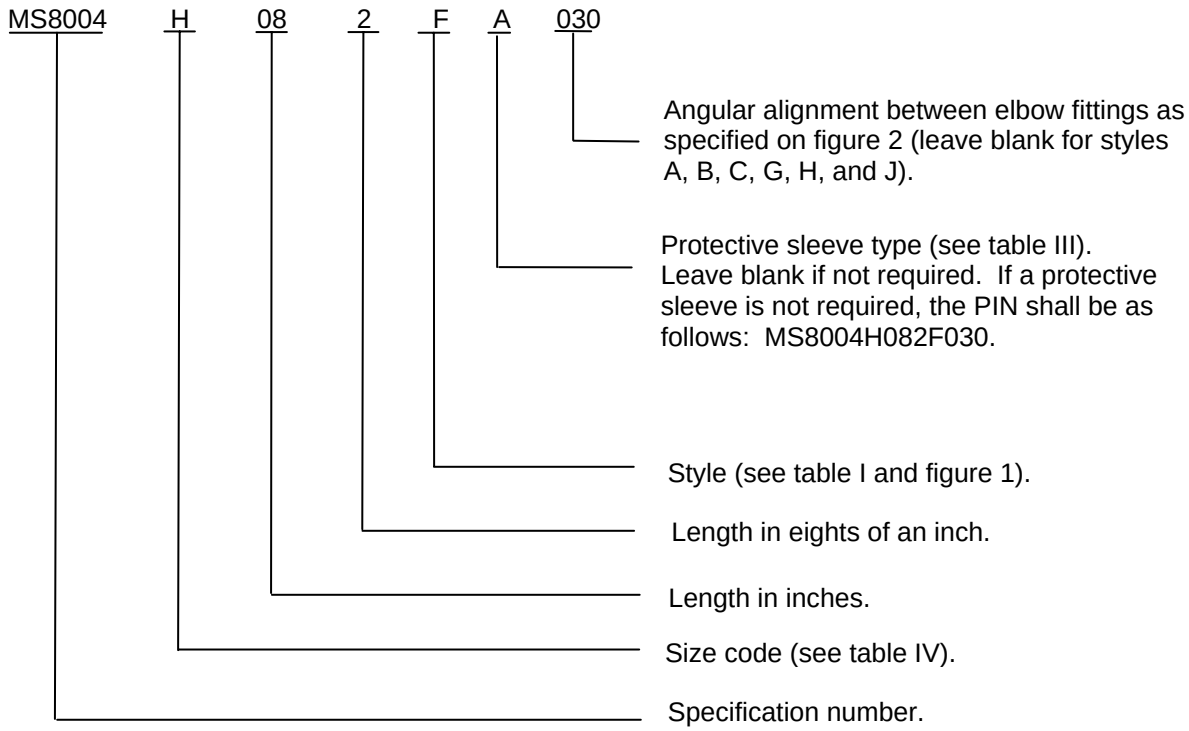
NOTES:
1. Hose assembly length “L” shall be measured, with the hose laid out horizontally and straight between the centers of the flange sealing surfaces, along a straight line parallel to the hose length.
2. See table IV for dimensions of (a) and (b).

FIGURE 3. Elbow fitting drop height, dimensions (a) and (b).

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PIN: The PIN for each hose assembly shall include its size code, length, style, protective sleeve type, and the angular alignment between the elbow fittings, as applicable.

Example: The PIN for an 8.250 inch (209.55 mm) length, style F hose assembly with a .500 inch (12.70 mm) tube OD, a fire protective sleeve in accordance with SAE AS1072, and a 30° between the elbow fittings shall be as follows:



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Referenced documents. In addition to MIL-DTL-25579, this document references the following:

MIL-DTL-27267
MIL-DTL-27272
SAE AS1073
SAE AS1291
SAE AS1298
SAE AS1072
MS27054
MS27056
MS27058
MS33786

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-0431-000)

Review activities:
Army - AR, AT, EA, MI
Navy - MC, SA, SH
Air Force – 70

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